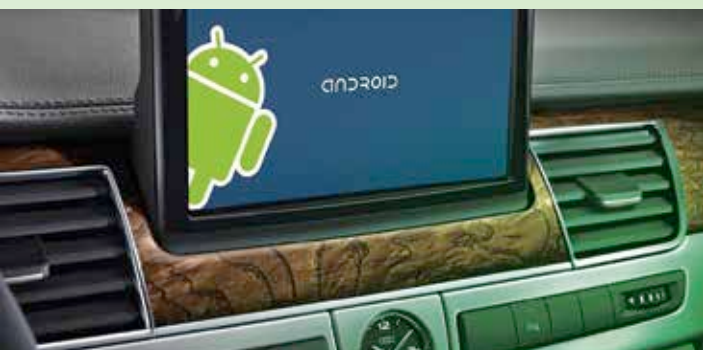


participants of the 'Android in cars' programme. Volvo's Sensus and Audi's MMI infotainment systems have been critically acclaimed in the industry, Volvo in particular have already been appreciated for the near-tablet like functionality it provides with its portrait-oriented infotainment system, making it the forerunner to present this technology.



You will soon get a full app store and an AI assistant in your car!



The Volvo Sensus interface will be there, but more like a skin

What is Android in Cars?

At Google I/O 2018, Volvo and Google introduced the future of Android in cars with a demonstration of the Volvo XC40. Android in cars essentially converts its tablet-like infotainment system into an Android tablet/smartphone.

The new system works independently of smartphones, and just like an Android device, you can simply log in to the system with your Google account. The features, though, include more than just media settings - you can actually store and synchronise all your memory settings to your Google account logged into the system.

You can gain access to all your saved preferences via the Google account, including saved locations, calendar entries, speed dial contacts

and more. This essentially helps in a more seamless usage scenario. The new system also gets integrated Google Maps, its own optimised Play Store and the Google Assistant, which powers all the voice actions.

The system is also open to third party developers, who can also build car-specific apps. Every option, from app interfaces and shortcut buttons can be custom-built, just like in any Android ecosystem. This ultimately aids versatility and helps with easier synchronisation of vehicle-to-vehicle connectivity.

How is it different?

With Android Auto, Google presented a considerably rudimentary way of getting Android OS in cars. It uses the device's wireless connections to mirror the phone's display to a compatible head unit - namely, car companies that fit a display larger than six inches, and subscribe to Google for its Android Auto services.

Once you set up Android Auto in your car, you can view Google Maps, attend handsfree calls, and have access to a host of other services. If you have local music on your phone, it will automatically sync under the Google Play Music tab. A few music streaming services are supported, but have to be downloaded to the phone.

With Android in cars, you can get all of this, and some more. To begin with, running Android directly in cars provides a uniform framework of code for all car manufacturers and developers to work with. This is exactly the same as the many custom UIs that smartphone manufacturers present on their devices.

Android in cars will completely do away with smartphone dependency, and have its own internet connection.

This will help in six key ways -

- 01.** Providing network for the always-on connected infrastructure
- 02.** Stores and analyses your driving data to automatically tune drivetrain settings
- 03.** Consult road and weather data to adjust drivetrain, select best routes and manage time of travel
- 04.** Use your Google account to learn your preferences, and suitably suggest music, podcasts, etc.
- 05.** Sync to your calendar and provide alerts for refuelling, servicing and any other car-based requirements
- 06.** Allow cars to communicate with each other and satellites to get real-time traffic information

With Android, what connected cars will also get is a full-fledged store of apps and a large developer base, and the list of apps being optimised for in-car usage is bound to grow.

Android is the future

We live in a mobile-first world, where all our settings and preferences are synchronised with our email accounts, and most of our times are spent accessing information of some kind through phone screens. With increasing congestion and the impending rise of in-car connectivity, it makes sense to have a smart-phone-independent operating unit in your car?

Android is the future of connected cars, and Volvo and Audi is possibly only the first patrons of yet another network that Google plans to dominate. With native apps (and a fully-stocked Play Store with every genre of apps), the integrated, AI-powered Google Assistant and standalone calling, messaging and emailing capabilities, Android is the perfect future from end-user perspective.

From the technology perspective too, Android in cars provides a readymade framework of servers for companies to build security protocols on. It can also help companies retain their custom interfaces, while providing the goodness of a complete App Store, full-fledged internet connectivity and a competent AI assistant. It is this that will form the immediate future for connected cars. ■